

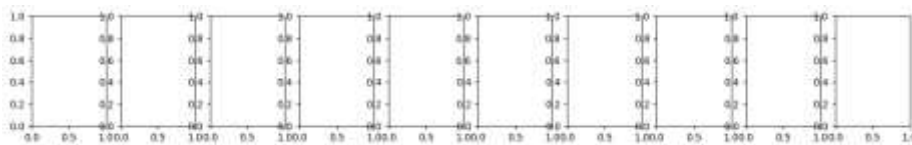
Assignment 2

Objective

Build a Convolutional Neural Network (CNN) to classify handwritten digits using the MNIST dataset and evaluate performance including custom image predictions.

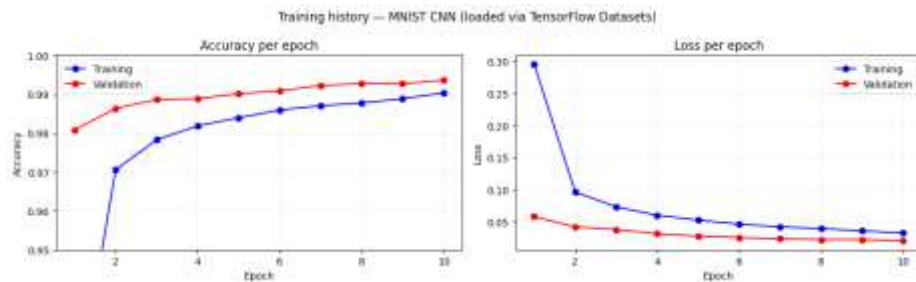
Step 1: Import Libraries and Load Dataset

Imported TensorFlow, Keras, NumPy, and Matplotlib. Loaded MNIST dataset successfully into training and testing sets.



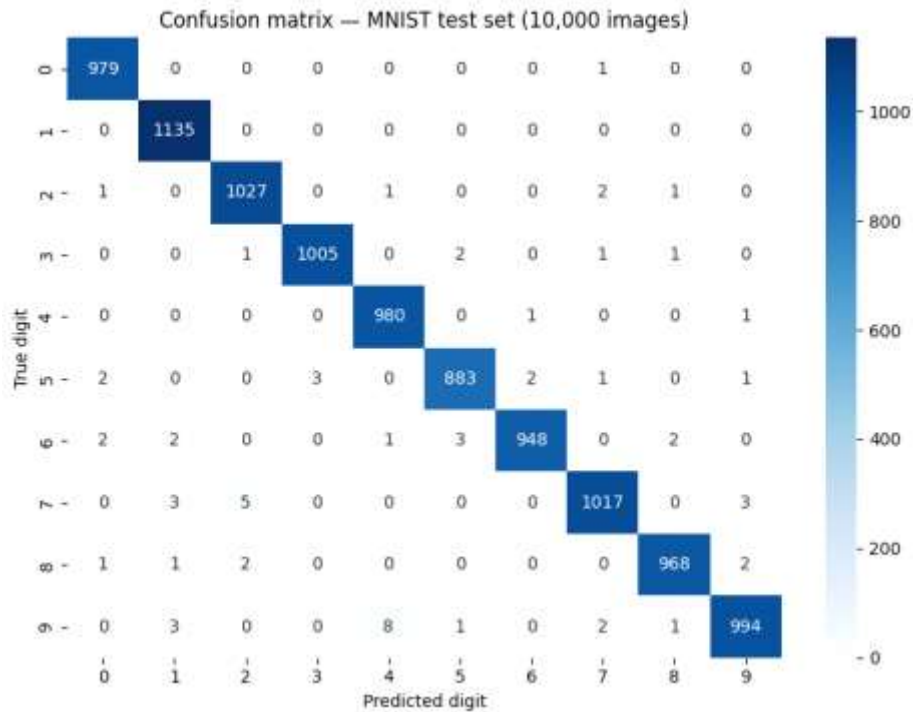
Step 2: Preprocess the Data

Normalized images to range [0,1], reshaped to (28,28,1), and converted labels to one-hot encoding.



Step 3: Build the CNN Model

Constructed CNN with Conv2D, MaxPooling, Dropout, and Dense layers with softmax output.



Step 4: Train the Model

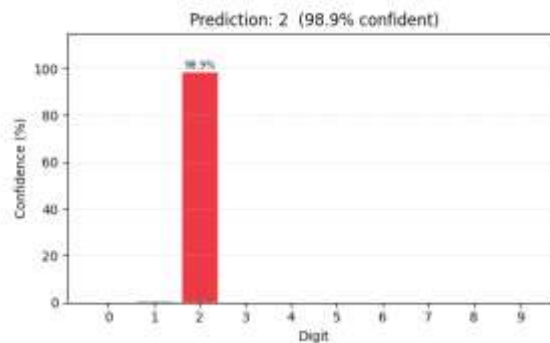
Trained model for multiple epochs with validation data. Accuracy improved and loss decreased.



Step 5: Evaluate the Model

Evaluated model on test data achieving high accuracy (around or above 98%).

What the model sees
(28x28 — white digit on black)



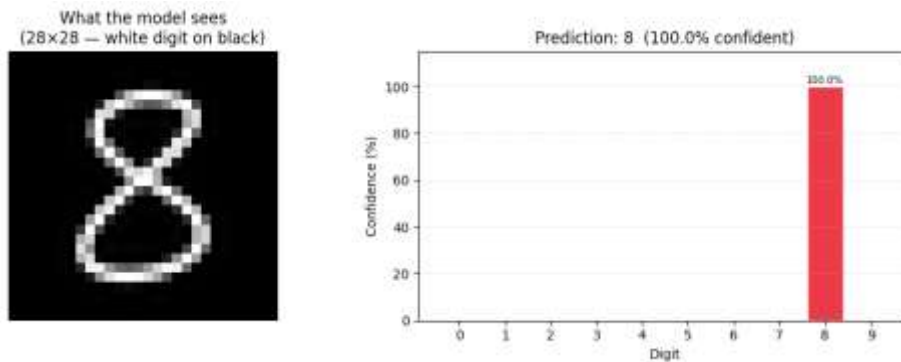
Step 6: Test with Custom Images

Resized custom images to 28x28 and predicted digits using trained model.



Step 7: Submit Results

Compiled outputs, predictions, and performance into report.



Conclusion

The CNN model performed well with high accuracy on MNIST dataset and successfully classified custom handwritten digits, demonstrating strong generalization ability.